Assignment 8

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Preliminaries

Package Loading

```
library(maptools)
library(stringr)
library(ggplot2)
library(readr)
library(dplyr, warn.conflicts=F)
library(ggmap)
library(maps)
library(mapdata)
library(rgeos)
library(sp)
library(rgdal)
library(lubridate)
library(ggrepel)
```

Assignment Part I

1) ***Answer to Problem 1***

Given by the quesiton, we have $var(X) = var(Y)$

By definition,

$$cor(X, Y) = \frac{cov(X, Y)}{\sqrt{var(X)var(Y)}}$$

If the two random variables are uncorrelated, the correlation will be 0 and the covariance would be 0.

$$\begin{aligned}
&cov(X - Y, X + Y) = E[(X - Y)(X + Y)] - E[X - Y]E[X + Y] \\
&= E[X^2 - Y^2] - (E[X]^2 - E[Y]^2) \\
&= E[X^2] - E[Y]^2 - (E[Y^2] - E[Y]^2) \\
&= var(X) - var(Y) \\
&= 0
\end{aligned}$$

So we have $X - Y$ and $X + Y$ are uncorrelated

2) ***Answer to Problem 2***

Given by the quesiton, we have $E(W) = E(X) = E[Y] = E[Z] = 0$, and $var(W) = var(X) = var[Y] = var[Z] = 1$, and W, X, Y, Z is pairwise uncorrelated. These imply that,

$$E[X^2] = var(x) + E[X]^2 = 1 + 0 = 1$$

Similarly, we have $E[Y^2] = E[W^2] = E[Z^2] = 1$

Thus, we have

a)

$$\begin{aligned}
&cov(R, S) = E[(W + X)(X + Y)] - E[W + X]E[X + Y] \\
&= E[WX + WY + X^2 + XY] - 0 \\
&= E[WX] + E[WY] + E[X^2] + E[XY] \\
&= E[X^2] \\
&= 1
\end{aligned}$$

and,

$$\begin{aligned}
&var(R) \\
&= var(W + X) \\
&= E[(W + X)^2] - E[(W + X)]^2 \\
&= E[(W^2 + X^2 + 2WX)] - 0 \\
&= 2
\end{aligned}$$

Similarly, we have $var(S) = var(T) = 2$

Thus we have,

$$\begin{aligned}
&\rho(R, S) = \frac{cov(R, S)}{\sqrt{var(R)var(S)}} \\
&= \frac{1}{\sqrt{2 \times 2}} \\
&= \frac{1}{2}
\end{aligned}$$

b)

$$\begin{aligned}
&cov(R, T) = E[(W + X)(Y + Z)] - E[W + X]]E[Y + Z] \\
&= E[WY + WZ + XY + XZ] - 0 \\
&= 0
\end{aligned}$$

and,

$$
\begin{aligned}
\rho(R,T) &= \frac{cov(R,T)}{\sqrt{var(R)var(T)}} \\
&= \frac{0}{\sqrt{2\times 2}} \\
&= 0
\end{aligned}
$$

Thus, the correlation coefficients $\rho(R,S) = \frac{1}{2}$ and $\rho(R,T) = 0$

3) ***Answer to Problem 3***

Given by the quesiton, we have $E(X) = 0$, $E(X^2) = 1$, $E[X^3] = 0$, $E[X^4] = 3$, These imply that

$$
\begin{aligned}
var(X) &= E[X^2] - E[X]^2 = 1 \\
var(X^2) &= E[X^4] - E[X^2]^2 = 2
\end{aligned}
$$

Thus,

$$
\begin{aligned}
&cov(X,Y) = E[XY] - E[X]E[Y] \\
&= E[aX + bX^2 + cx^3] - E[X]E[a + bX + xX^2] \\
&= aE[X] + bE[b^2] + cE[X^3] - E[X](a + bE[X] + cE[X^2]) \\
&= 0 + b + 0 - 0 \\
&= b
\end{aligned}
$$

We also have $var(X) = E[X^2] - E[X]^2 = 1$

and,

$$
\begin{aligned}
&var(Y) = var(a + bX + cX^2) \\
&= b^2 var(X) + c^2 var(X^2) + 2cov(X,X^2) \\
&= b^2 + 2c^2 + 2(E[X^3] - E[X]E[X^2]) \\
&= b^2 + 2c^2 + 0 \\
&= b^2 + 2c^2
\end{aligned}
$$

Thus,

$$
\begin{aligned}
\rho(X,Y) &= \frac{cov(X,Y)}{\sqrt{var(X)var(Y)}} \\
&= \frac{b}{\sqrt{b^2 + 2c^2}}
\end{aligned}
$$

4) ***Answer to Problem 4***

Given by the question, we have X~$Normal(0,1)$, and its PDF is $f_X(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$

The corresponding transform is given by

$$
\begin{aligned}
&M(t)\\
&= E[e^{tX}]\\
&= \int_{\forall x} \left(e^{tx} f_X(x)\right) dx\\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{x^2}{2}} dx\\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx-\frac{x^2}{2}} dx\\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{x^2-2tx+t^2}{2}} e^{\frac{t^2}{2}} dx\\
&= e^{\frac{t^2}{2}} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-t)^2}{2}} dx\\
&= e^{\frac{t^2}{2}}
\end{aligned}
$$

$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}e^{-\frac{(x-t)^2}{2}} dx = 1$ because it is constrained by the normalization condition.

Then, by $E[X^n] = \int_{-\infty}^{\infty} x^n f_X(x) dx = \frac{d^n}{dt^n} M(t)\Big|_{t=0}$, we have

$$
\begin{aligned}
E[X^3] &= \frac{d^3}{dt^3} M(t)\Big|_{t=0}\\
&= e^{\frac{t^2}{2}}(t^3 + 3t)\Big|_{t=0}\\
&= 0
\end{aligned}
$$

and,

$$
\begin{aligned}
E[X^4] &= \frac{d^4}{dt^4} M(t)\Big|_{t=0}\\
&= e^{\frac{t^2}{2}}(t^4 + 6t^2 + 3)\Big|_{t=0}\\
&= 3
\end{aligned}
$$

5) ***Answer to Problem 5***

Given by the question, we have the pdf $f_X(x) = \lambda e^{-\lambda x}$, $x \geqslant 0$

The corresponding transform is given by

$$
\begin{aligned}
&M(t)\\
&=\int_0^\infty \Big(e^{tx}f_X(x)\Big)dx\\
&=\lambda\int_0^\infty e^{tx}e^{-\lambda x}dx\\
&=\lambda\int_0^\infty e^{(t-\lambda)x}dx\\
&=\lambda\frac{e^(t-\lambda)x}{t-\lambda}\Big|_0^\infty \text{ if } t<\lambda\\
&=\lambda(\frac{0}{t-\lambda}-\frac{1}{t-\lambda})\\
&=\frac{\lambda}{\lambda-t}
\end{aligned}
$$

Thus we know that

$$
\begin{aligned}
\frac{d}{dt}M(t) &= \frac{\lambda}{(\lambda-t)^2}\\
\frac{d^2}{dt^2}M(t) &= \frac{2\lambda}{(\lambda-t)^3}\\
\frac{d^3}{dt^3}M(t) &= \frac{6\lambda}{(\lambda-t)^4}\\
\frac{d^4}{dt^4}M(t) &= \frac{24\lambda}{(\lambda-t)^5}\\
\frac{d^5}{dt^5}M(t) &= \frac{120\lambda}{(\lambda-t)^6}
\end{aligned}
$$

Finally, we get that

$$
\begin{aligned}
E[X^3] &= \frac{d^3}{dt^3}M(t)\Big|_{t=0}\\
&= \frac{6\lambda}{(\lambda-t)^4}\Big|_{t=0}\\
&= 6\lambda^{-3}
\end{aligned}
$$

and,

$$
\begin{aligned}
E[X^4] &= \frac{d^4}{dt^4}M(t)\Big|_{t=0}\\
&= \frac{24\lambda}{(\lambda-t)^5}\Big|_{t=0}\\
&= 24\lambda^{-4}
\end{aligned}
$$

and,

$$
\begin{aligned}
E[X^5] &= \frac{d^5}{dt^5} M(t)\Big|_{t=0} \\
&= \frac{120\lambda}{(\lambda - t)^6}\Big|_{t=0} \\
&= 120\lambda^{-5}
\end{aligned}
$$

Assignment Part II

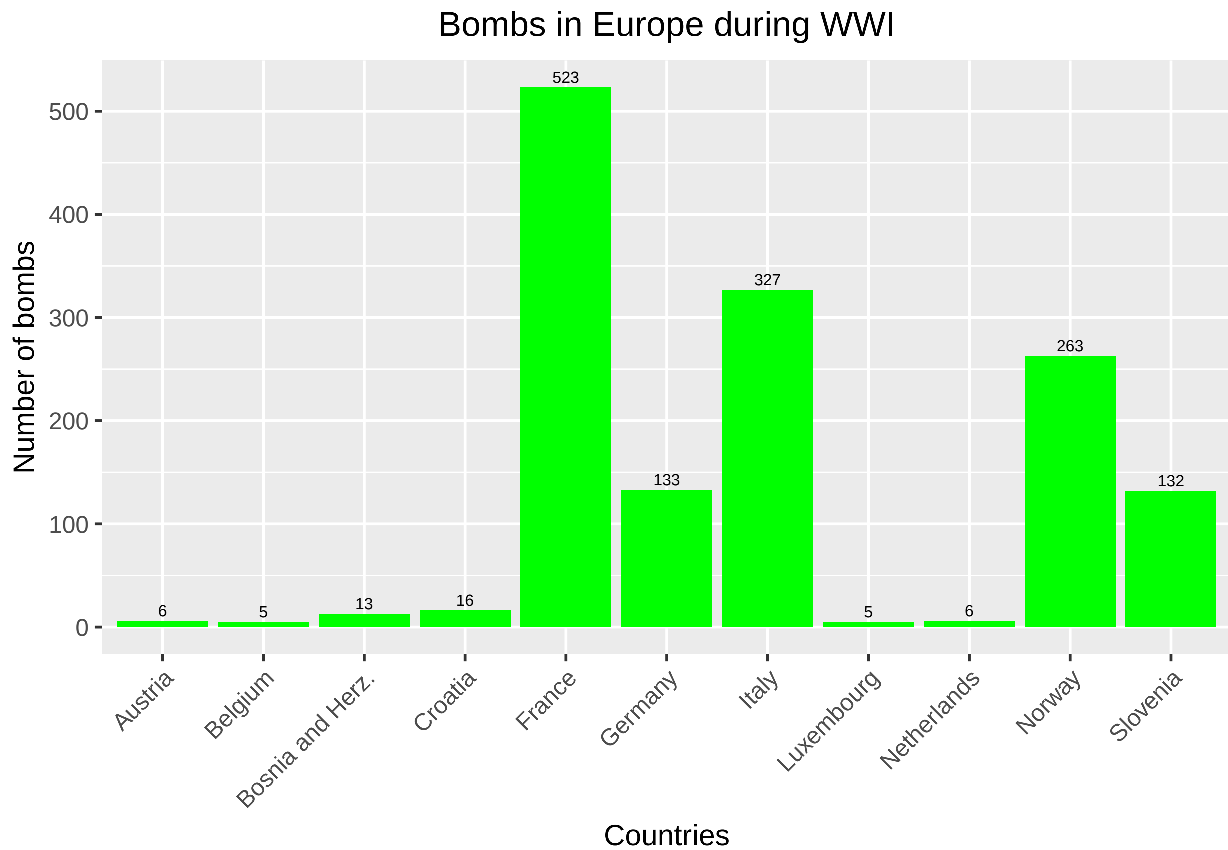
6) ***Answer to Problem 6***

Get the data,

```
shape1 <- readShapeSpatial("~/Dropbox/courses/Mathematic Fundations of Political Methodology/Weekly R la
shape1@data$id<-rownames(shape1@data) #make a data.frame out of the shape file, therefore we need some 

# making a variable called region from the id variable
shape.fort<-fortify(shape1, region="id")

shape.data<-shape.fort%>%left_join(data.frame(shape1))

df<-read_csv("~/Dropbox/courses/Mathematic Fundations of Political Methodology/Weekly R lab/week 8/THOR
```

Then plot

```
tableEurope <- function(x) {
  europefile <- shape.data[shape.data$NAME == x, ]
  bombsin <- point.in.polygon(point.x = as.numeric(df$Lon), point.y = as.numeric(df$Lat),
                              pol.x = europefile$long, pol.y = europefile$lat)
  table(bombsin)["1"]
}

countries <- unique(unlist(shape.data$NAME))

bombsEurope1 <- mapply(countries, FUN = tableEurope)
bE1 <- as.data.frame(bombsEurope1)
bE1$names <- countries
bE2 <- na.omit(bE1)

bombsplot1 <- ggplot(bE2, aes(x=names, y=bombsEurope1)) + geom_bar(stat="identity", fill="green")

bombsplot1 + ggtitle ("Bombs in Europe during WWI")+
  theme(plot.title=element_text(hjust=0.5))+
  theme(axis.text.x = element_text(angle=45, hjust=1))+
  labs(x="Countries", y="Number of bombs")+
  geom_text(aes(label=bombsEurope1), size = 2, vjust=-0.4)
```



7) ***Answer to Problem 7***

The center of gravity is latitude=49.08321 and longtitude=6.47925.

I use bombload as the weight to calculate the center of gravity. The reason is that I think when considering the the degree of bombing, we need to cosider the load of bombs as well as the frequency of bombing. For instance, even if one place only experienced one bombing, if many bombs were dropped there, we still think the severity of bombing at this place is high.

I also calculate and plot the rolling average

```
df2<-df%>%mutate(Date2=as.Date(MSNDate, "%m/%d/%Y"))%>%mutate(month=month(Date2))%>%mutate(year=year(Da

gravitydt <- select(df2, c(Bombload, Date2, month, year, Lat, Lon, Country1))
gravitydt <- na.omit(gravitydt)

gravitydt2<-gravitydt%>%
  mutate(Lat=as.numeric(Lat))%>%
  mutate(bombload=as.numeric(Bombload))%>%
  mutate(Lon=as.numeric(Lon))%>%
  na.omit()

gravitydt2$aveblat <- (sum(gravitydt2$Lat*gravitydt2$bombload)/sum(gravitydt2$bombload))
gravitydt2$aveblon <- (sum(gravitydt2$Lon*gravitydt2$bombload)/sum(gravitydt2$bombload))

gravitydt2$aveblat[1]
```

```
## [1] 49.08321
```

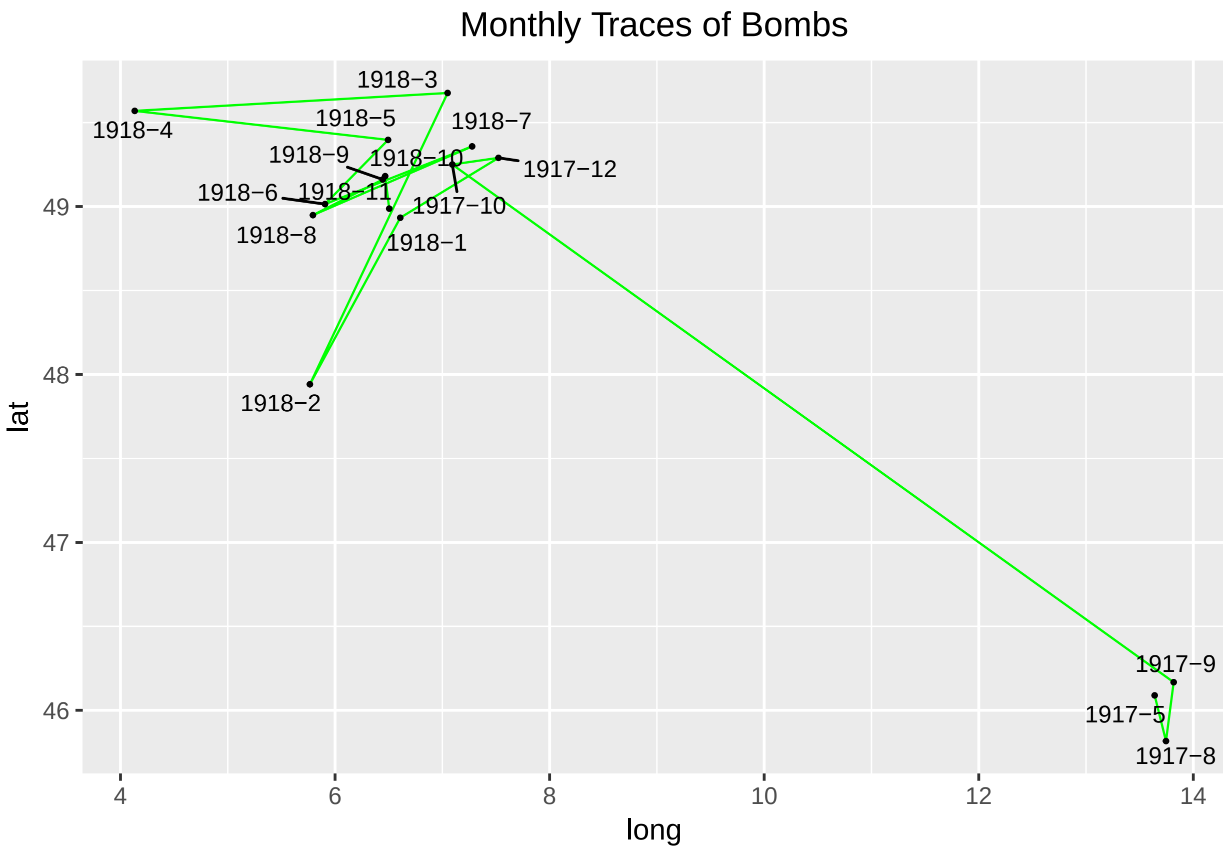
```r
gravitydt2$aveblon[1]
```

```
## [1] 6.47925
```

```r
#Monthly Rolling Average
gravitydt4<-gravitydt2%>%
  group_by(month, year)%>%
  summarise(rollavg1= sum(Lat*bombload)/sum(bombload),rollavg2= sum(Lon*bombload)/sum(bombload))

gravitydt4 <- gravitydt4[order(gravitydt4$year), ]
gravitydt4
```

```
## Source: local data frame [17 x 4]
## Groups: month [12]
##
##    month  year rollavg1  rollavg2
##    <dbl> <dbl>    <dbl>     <dbl>
## 1      9  1916 45.85100 13.584000
## 2      5  1917 46.08860 13.639700
## 3      8  1917 45.81663 13.745310
## 4      9  1917 46.16700 13.817000
## 5     10  1917 49.25057  7.092896
## 6     12  1917 49.29006  7.523319
## 7      1  1918 48.93351  6.607973
## 8      2  1918 47.94142  5.765356
## 9      3  1918 49.67648  7.049206
## 10     4  1918 49.56982  4.132581
## 11     5  1918 49.39707  6.494250
## 12     6  1918 49.01451  5.906304
## 13     7  1918 49.35847  7.277833
## 14     8  1918 48.94885  5.793793
## 15     9  1918 49.16113  6.444817
## 16    10  1918 49.18103  6.467168
## 17    11  1918 48.98786  6.505020
```

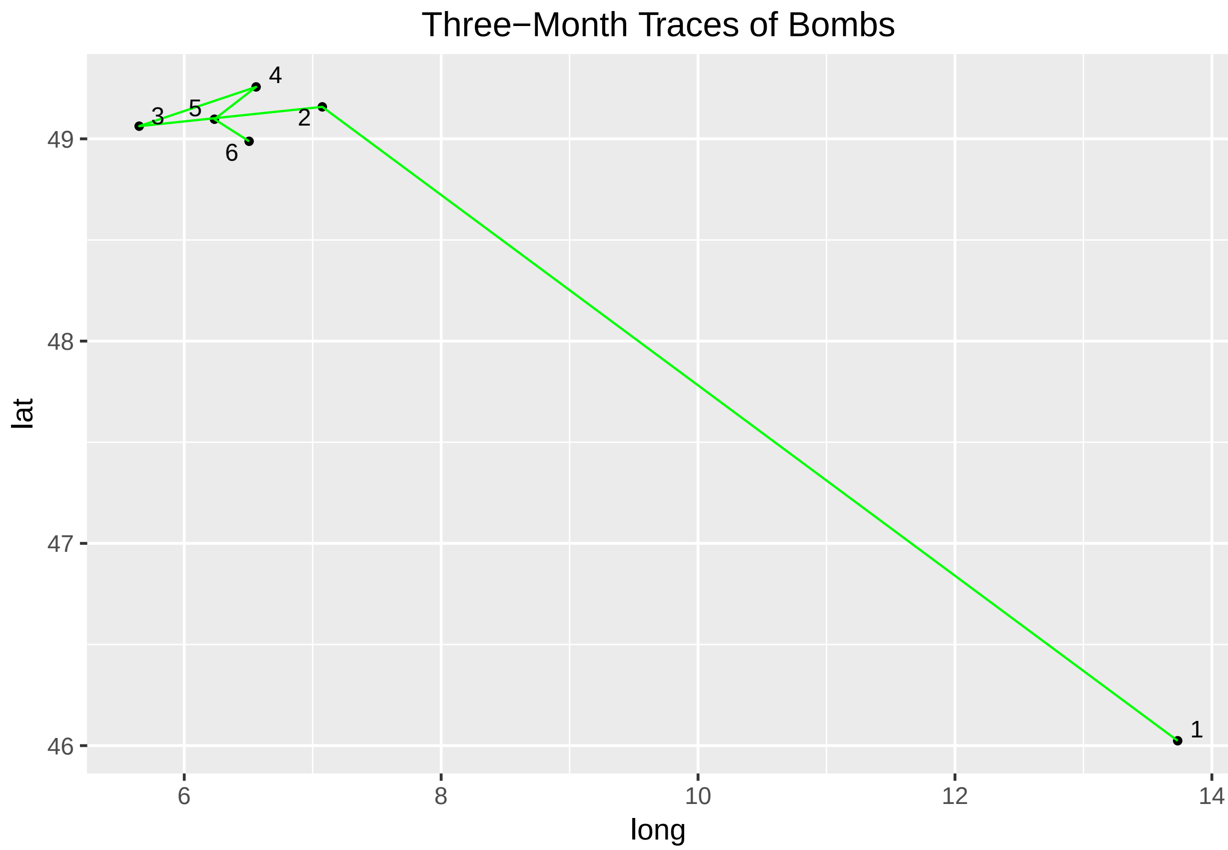
```r
#Plotting Monthly Rolling Average
gravitydt4$YearMonth <- paste(gravitydt4$year, gravitydt4$month, sep="-" )

gravitydt4_1 <- gravitydt4[-c(1), ]

bombstrace1_1 <- ggplot(gravitydt4_1, aes(rollavg2, rollavg1)) +
  ggtitle ("Monthly Traces of Bombs") +
  geom_path(size=.4, colour="green") +
  guides(color=guide_legend(override.aes= list(size=3))) +
  theme(plot.title=element_text(hjust=0.5)) + geom_point(size=.5)
bombstrace1_1 +labs(x="long", y="lat") +
 geom_text_repel(aes(rollavg2, rollavg1, label=YearMonth), size=3)
```




```r
#Three-month rolling average
gravitydt4_1 <- gravitydt4[-c(1), ]
gravitydt4_1$tri <- c(1,1,1,2,2,2,3,3,3,4,4,4,5,5,5,6)
gravitydt5_1 <-gravitydt4_1%>%group_by(tri)%>%summarise(rollavg3 = mean(rollavg1), rollavg4 = mean(rolla
gravitydt5_1
```

```
## # A tibble: 6 × 3
##     tri rollavg3  rollavg4
##   <dbl>    <dbl>     <dbl>
## 1     1 46.02408 13.734003
## 2     2 49.15805  7.074730
## 3     3 49.06257  5.649048
## 4     4 49.25669  6.559462
## 5     5 49.09700  6.235259
## 6     6 48.98786  6.505020
```

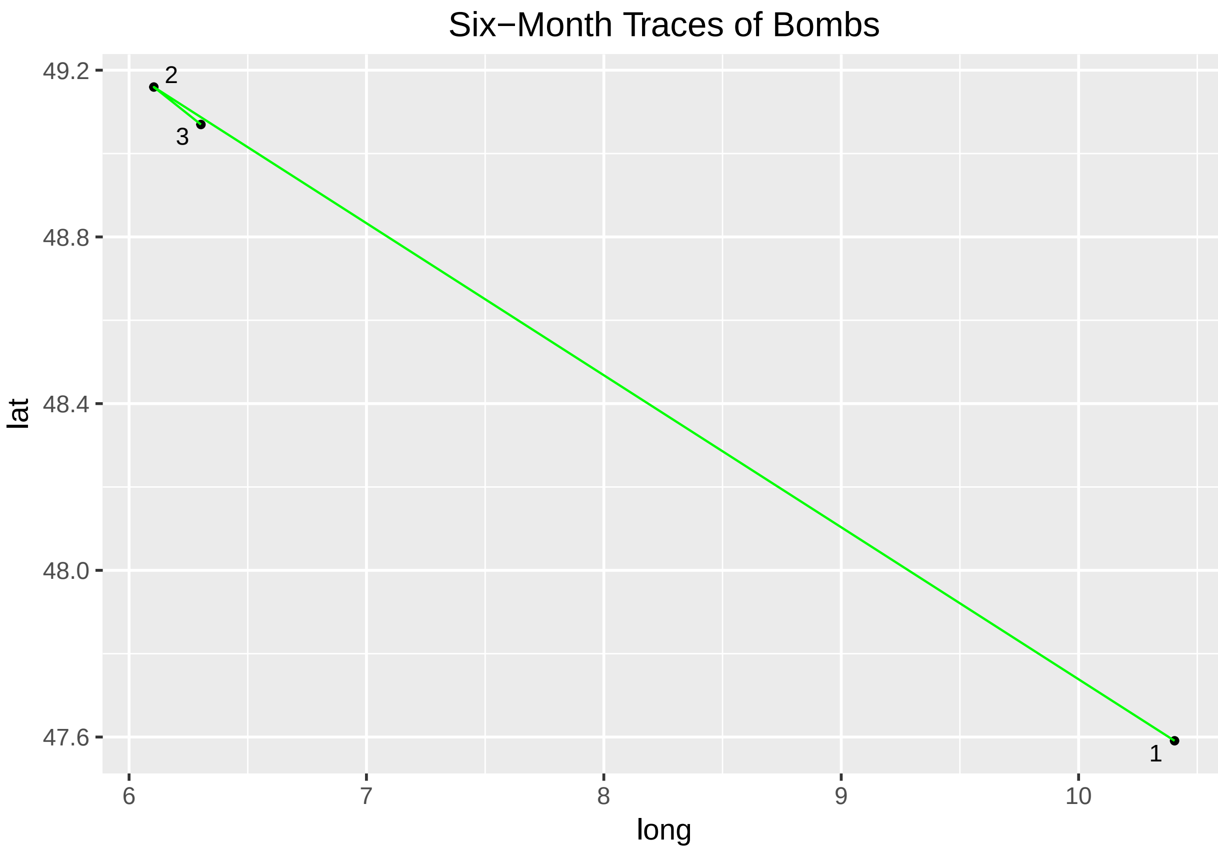
```r
#Plotting tri-monthly rolling Average
bombstrace2_1 <- ggplot(gravitydt5_1, aes(rollavg4, rollavg3)) +ggtitle ("Three-Month Traces of Bombs")
theme(plot.title=element_text(hjust=0.5)) +  geom_path(size=.4, colour="green") +
  guides(color=guide_legend(override.aes= list(size=3)))
bombstrace2_1 +labs(x="long", y="lat") + geom_text_repel(aes(rollavg4, rollavg3, label=tri), size=3)
```



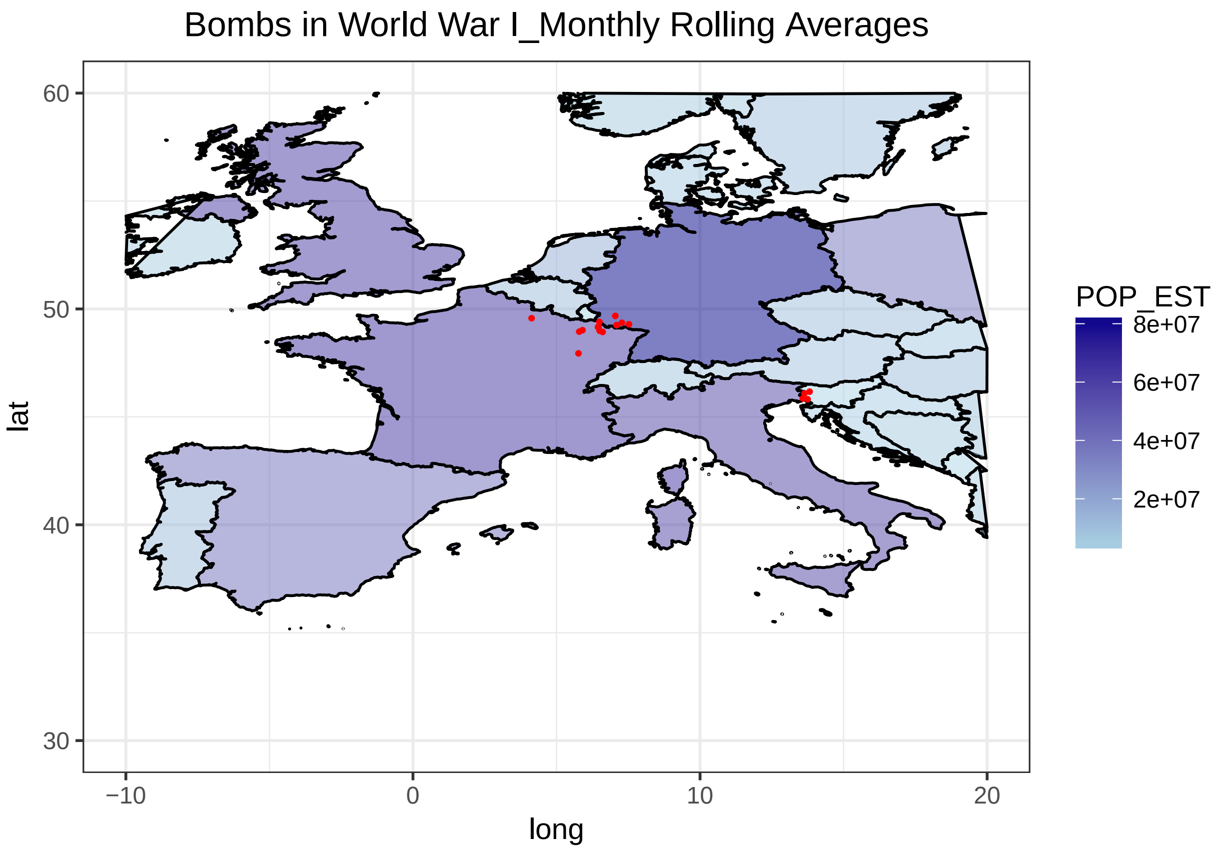

```
#Six-month rolling average
gravitydt4_1 <- gravitydt4[-c(1), ]
gravitydt4_1$six <- c(1,1,1,1,1,1,2,2,2,2,2,2,3,3,3,3)
gravitydt6_1 <-gravitydt4_1%>%group_by(six)%>%summarise(rollavg5 = mean(rollavg1),
rollavg6 = mean(rollavg2))
gravitydt6_1
```

```
## # A tibble: 3 × 3
##     six rollavg5  rollavg6
##   <dbl>    <dbl>     <dbl>
## 1     1 47.59106 10.404366
## 2     2 49.15963  6.104255
## 3     3 49.06972  6.302700
```

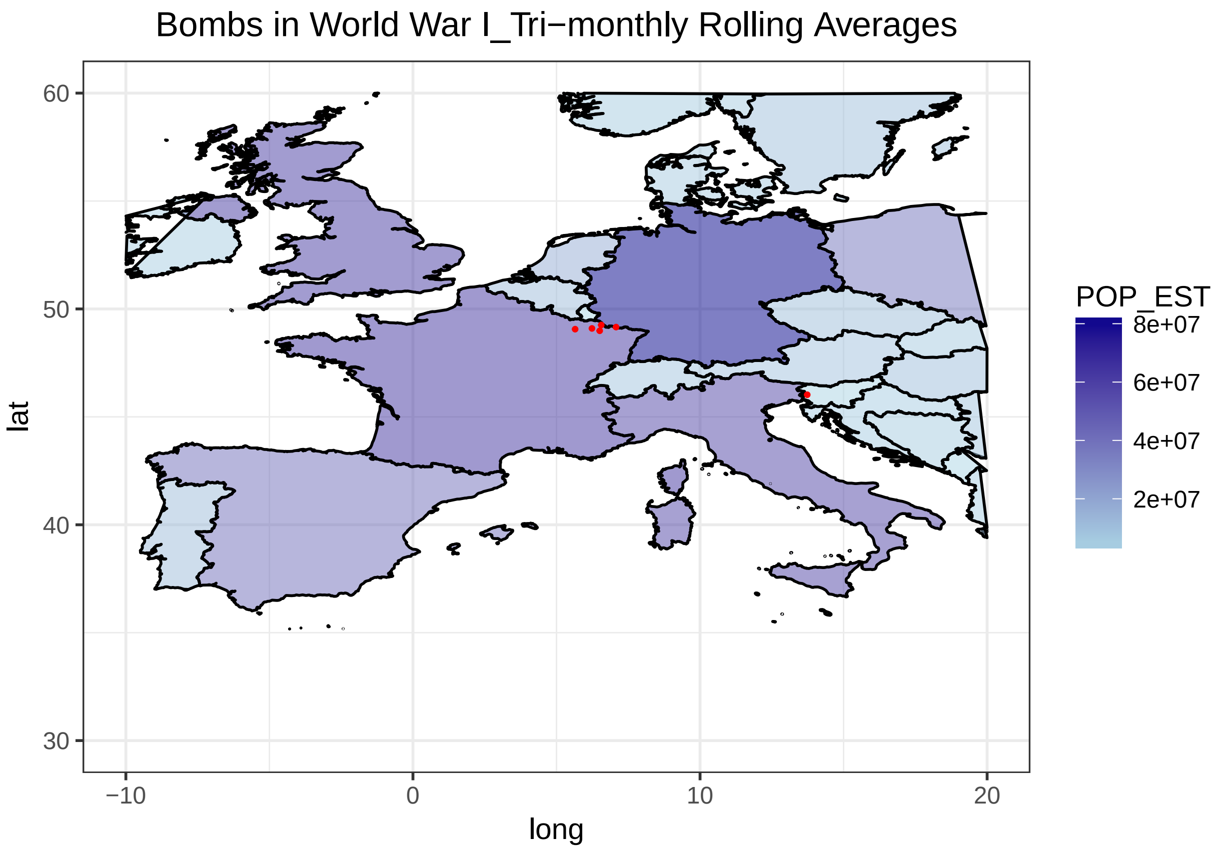
```
#Plotting six-monthly rolling average
bombstrace3_1 <- ggplot(gravitydt6_1, aes(rollavg6, rollavg5)) +ggtitle ("Six-Month Traces of Bombs") +
geom_point(size=1) + guides(color=guide_legend(override.aes= list(size=3))) +
theme(plot.title=element_text(hjust=0.5)) + geom_path(size=.4, colour="green") +
guides(color=guide_legend(override.aes= list(size=3)))
bombstrace3_1 +labs(x="long", y="lat") + geom_text_repel(aes(rollavg6, rollavg5, label=six), size=3)
```



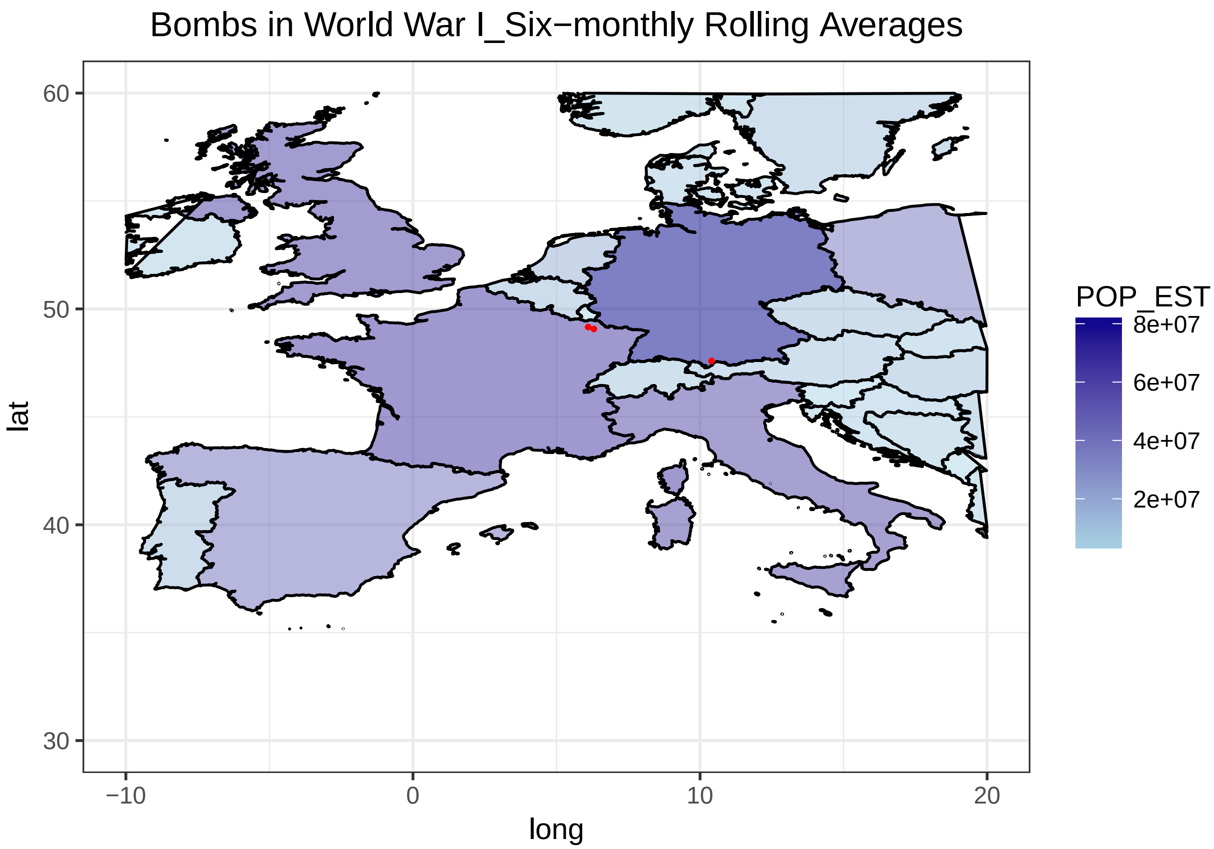

```r
#The location of the monthly rolling averages in real map
ggplot() +
  geom_polygon(data = shape.data, colour = "black",
               aes(long, lat, group = group, fill = POP_EST), alpha = 0.5) +
  scale_fill_gradient(low = "lightblue", high = "darkblue", na.value = "lightyellow") +
  geom_point(data = gravitydt4, aes(x=rollavg2, y=rollavg1), size = .5, colour="red") +
  xlim(-10, 20) + ylim(30, 60) + theme_bw() +
  ggtitle("Bombs in World War I_Monthly Rolling Averages") +theme(plot.title=element_text(hjust=0.5))
```




```
#The location of the tri-monthly rolling averages in real map
ggplot() +
  geom_polygon(data = shape.data, colour = "black",
               aes(long, lat, group = group, fill = POP_EST), alpha = 0.5) +
  scale_fill_gradient(low = "lightblue", high = "darkblue", na.value = "lightyellow") +
  geom_point(data = gravitydt5_1, aes(x=rollavg4, y=rollavg3), size = .5, colour="red") +
  xlim(-10, 20) + ylim(30, 60) + theme_bw() +
  ggtitle("Bombs in World War I_Tri-monthly Rolling Averages") +
  theme(plot.title=element_text(hjust=0.5))
```




```r
#The location of the six-month rolling averages in real map
ggplot() +
  geom_polygon(data = shape.data, colour = "black",
               aes(long, lat, group = group, fill = POP_EST), alpha = 0.5) +
  scale_fill_gradient(low = "lightblue", high = "darkblue", na.value = "lightyellow") +
  geom_point(data = gravitydt6_1, aes(x=rollavg6, y=rollavg5), size = .5, colour="red") +
  xlim(-10, 20) + ylim(30, 60) + theme_bw() +
  ggtitle("Bombs in World War I_Six-monthly Rolling Averages")+
  theme(plot.title=element_text(hjust=0.5))
```


Bombs in World War I_Six−monthly Rolling Averages
60
50
40
30
lat
−10
0
10
20
long
POP_EST
8e+07
6e+07
4e+07
2e+07